

Lab10

LAB 10

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PDB STATS

```
pdb_data <- read.csv("pdb_stats.csv")
pdb_data
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	214078
2	1	0	13981
3	0	0	15759
4	3	1	4941
5	0	0	237
6	0	4	22

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

Answer Q1: The percentage of structures in the PDB that are solved by X-ray and Electron Microscopy is 93.7892%.

```
xray_total <- sum(pdb_data$X.ray)

em_total <- sum(pdb_data$EM)

all_total <- sum(pdb_data$Total)

percent_combined <- ((xray_total + em_total)/all_total)*100
percent_combined
```

```
[1] 93.7892
```

Q2: What proportion of structures in the PDB are protein?

Answer Q2: The proportion of structures in the PDB that are protein is 97.9118%.

```
rownames(pdb_data) <- pdb_data$Molecular.Type
protein_only <- pdb_data["Protein (only)", "Total"]

protein_oligosaccharide <- pdb_data["Protein/Oligosaccharide", "Total"]

protein_na <- pdb_data["Protein/NA", "Total"]

percent_protein <- (protein_only + protein_oligosaccharide + protein_na)/all_total *100
percent_protein
```

```
[1] 97.9118
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Answer Q3: When typing in HIV into the PDB website, and specifying a structure attribute search, selecting structure title and adding “HIV-1 protease” to find that there are 718 HIV-1 protease structures in the current PDB.

HIV-Pr ### Add a spacefill for Ligand and add a residue property to Polymer.

Cleaning up the display

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Answer Q4: We only see 1 atom per water molecule in this sticture because we are using the X-ray crystal structure of HIV-1 protease, meaning that only electrons in oxygen atoms are visible. That is beacuse X-rays scatter off electrons, and hydrogen have only 1 electron which is too weak of a signal. Therefore, water is recorded just as the oxygen atom and shows 1 atom per water molecule.

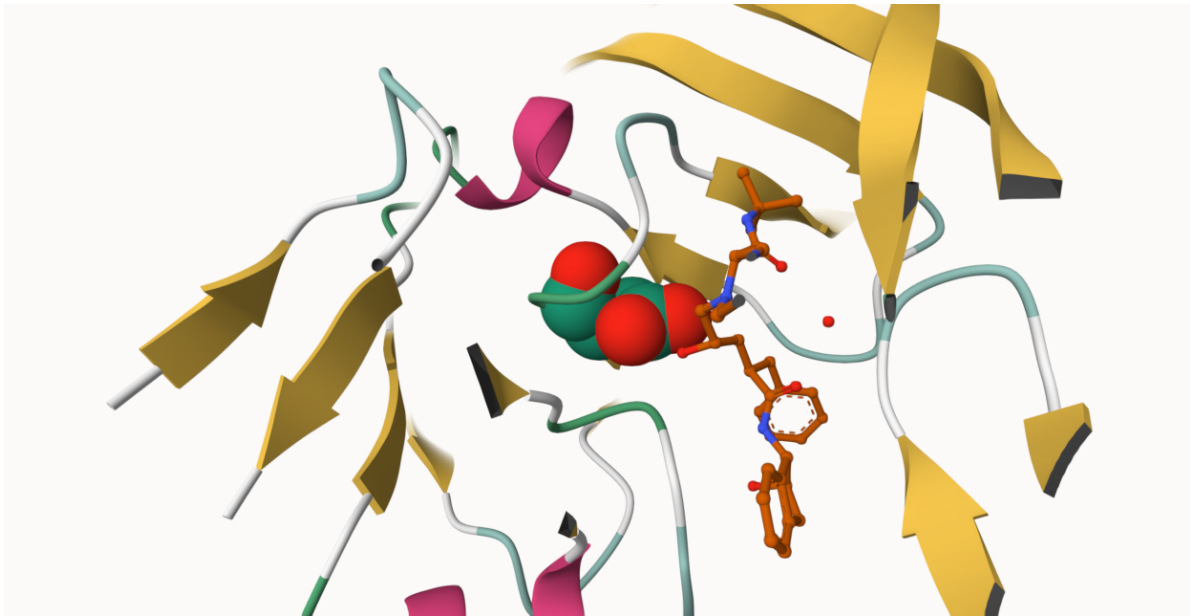
Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

Answer Q5: The critical conserved water molecule in the binding site is hydrogen bond between the ligand (positioned between critical residue)and it’s residue number is H308.

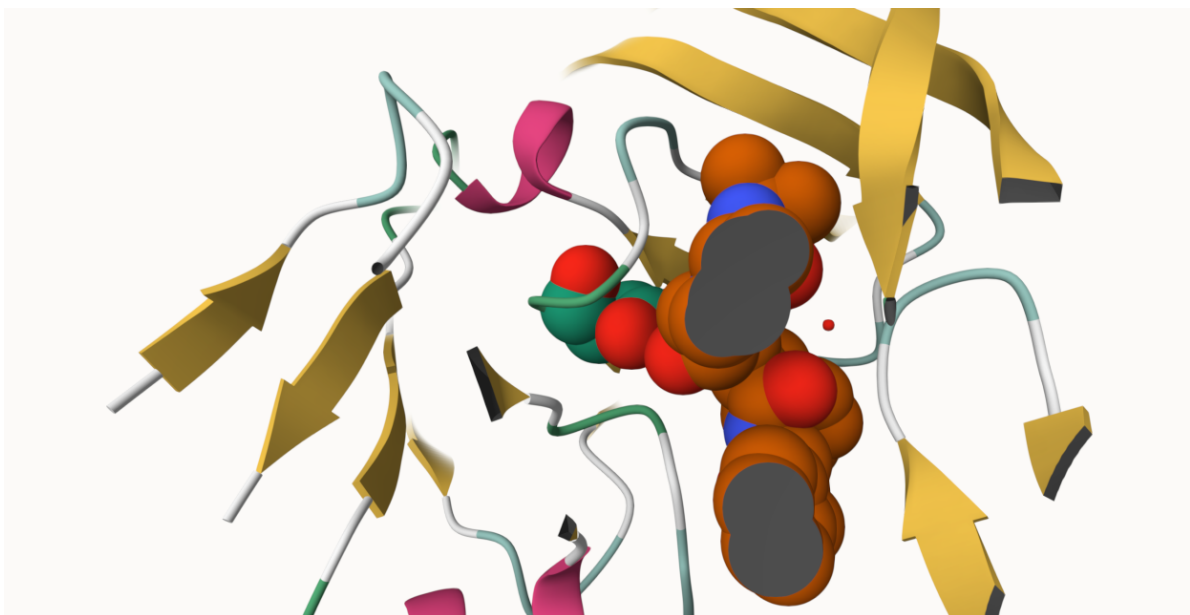
Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

Answer Q6: The images below show the ball and stick image and the spacefill image.

```
#Ball and Stick Image
knitr::include_graphics("1HSG 2.png")
```



```
#Spacefill Image  
knitr::include_graphics("1HSG1.png")
```



Introduction to Bio3D in R

```
#Accessing on-line PDB file
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
#Summary of pdb using print() function
print(pdb)
```

Call: read.pdb(file = "1hsg")

```
Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

Answer Q7: There are 198 amino acid (protein) residues in this pdb object.

Q8: Name one of the two non-protein residues?

Answer Q8: One of the two non-protein residues is HOH(127).

Q9: How many protein chains are in this structure?

Answer Q9: There are 2 protein chains in the structure.

Predicting functional motions of a single structure

```
#Read PDB structure of adk
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM TAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

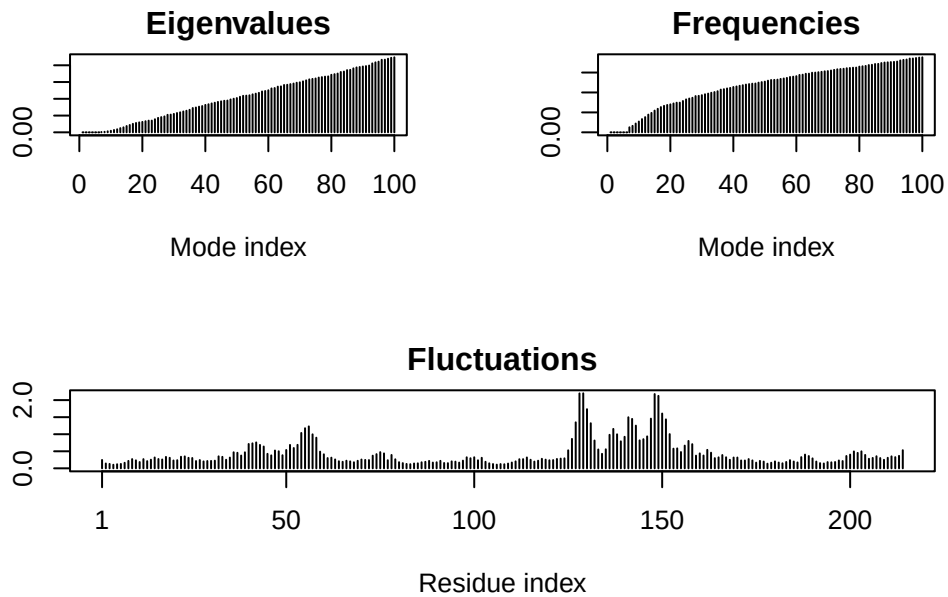
```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
# Perform flexibility prediction
m <- nma(adk)
```

```
Building Hessian...      Done in 0.036 seconds.
Diagonalizing Hessian... Done in 0.371 seconds.
```

plot m and view “movie” of these predicted motions

```
plot(m)
```



Comparative structure analysis of Adenylate Kinase

Setup

Q10. Which of the packages above is found only on BioConductor and not CRAN?

Answer Q10: The package “msa” is the only one found in BioConductor but not CRAN. It is distributed through BioConductor only and installed via “Bico-Manger”.

Q11. Which of the above packages is not found on BioConductor or CRAN?

Answer Q11: The package not found on BioConductor or CRAN is “bio3dview” as it is only installed through github.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

Answer Q12: True, the functions from the pak package can be used to install packages from GitHub and BitBucket. That is because pak package is used as a replacement for older installations (including CRAN,BioConductor,GitHub, Bitbucket).

Search and Retrieve ADK structures

```
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```


Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

Answer Q13: There are 214 amino acids in this sequence, as can be seen as there are “0 gap” meaning no insertions or deletions affect the length. So the “214 position columns” are the full sequence length.

Hits

```
hits <- NULL
hits$pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6HAP_A')

# Download related PDB files
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download
```

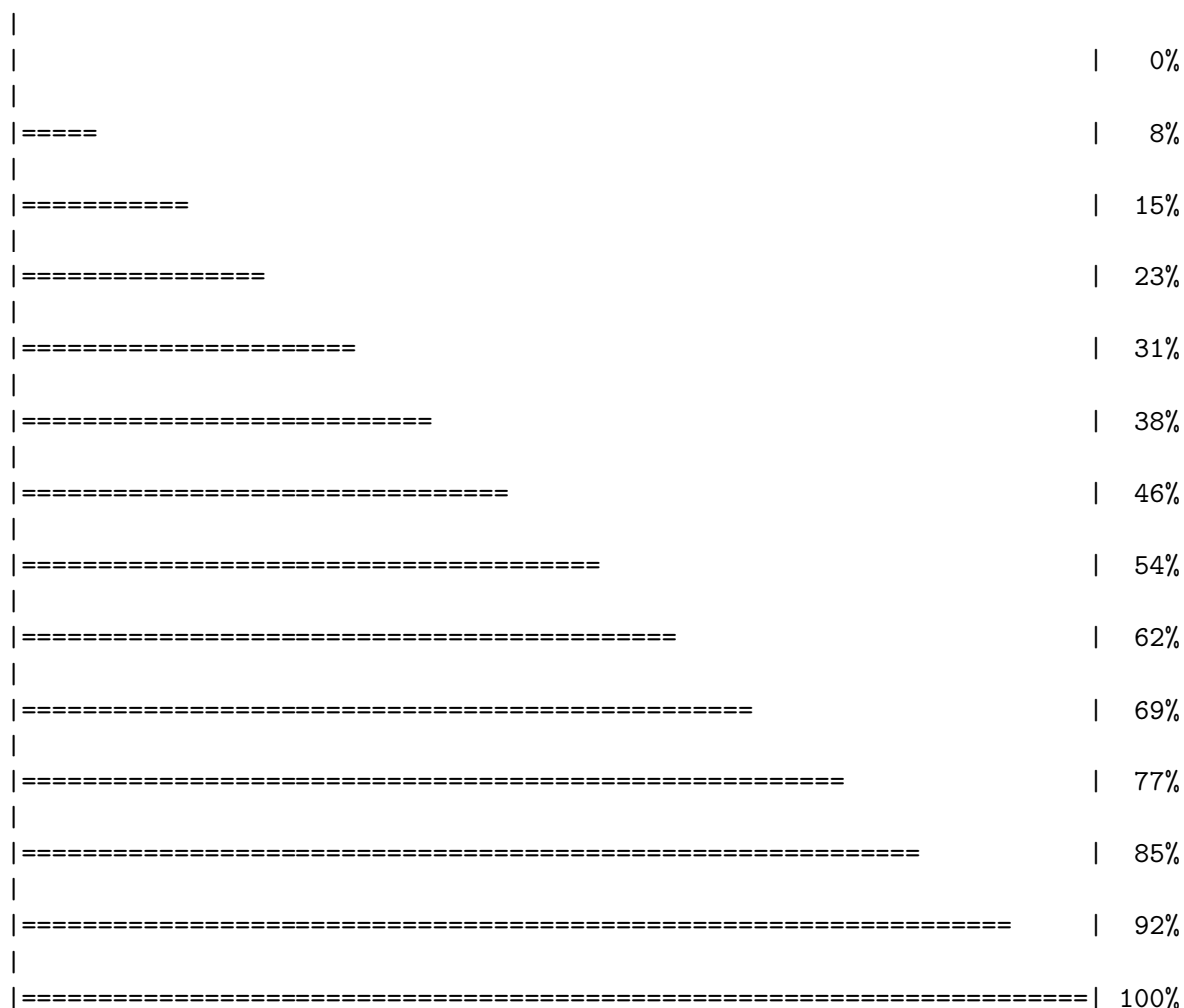
```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download



Align and Superpose structures

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...
```

Extracting sequences

```
pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
```

```

pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbs/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbs/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbs/split_chain/4PZL_A.pdb

```

Annotate collected PDB structures

```

# Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)

```

```

[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"

```

```

#View all available annotation data:
anno

```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-ray
6S36_A	6S36	A	Protein	214	X-ray
6RZE_A	6RZE	A	Protein	214	X-ray
3HPR_A	3HPR	A	Protein	214	X-ray
1E4V_A	1E4V	A	Protein	214	X-ray
5EJE_A	5EJE	A	Protein	214	X-ray
1E4Y_A	1E4Y	A	Protein	214	X-ray
3X2S_A	3X2S	A	Protein	214	X-ray
6HAP_A	6HAP	A	Protein	214	X-ray
6HAM_A	6HAM	A	Protein	214	X-ray
4K46_A	4K46	A	Protein	214	X-ray
3GMT_A	3GMT	A	Protein	230	X-ray

4PZL_A	4PZL	A	Protein	242	X-ray
	resolution	scopDomain			pfam
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
3X2S_A	2.80	<NA>			<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)		
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3GMT_A	2.10	<NA>			<NA>
4PZL_A	2.10	<NA>		Adenylate kinase (ADK)	
		ligandId			
1AKE_A		AP5			
6S36_A	CL (3),NA,MG (2)				
6RZE_A	NA (3),CL (2)				
3HPR_A		AP5			
1E4V_A		AP5			
5EJE_A		AP5,CO			
1E4Y_A		AP5			
3X2S_A	JPY (2),AP5,MG				
6HAP_A		AP5			
6HAM_A		AP5			
4K46_A	ADP,AMP,PO4				
3GMT_A	SO4 (2)				
4PZL_A	CA,FMT,GOL				
				ligandName	
1AKE_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE	
6S36_A				CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)	
6RZE_A				SODIUM ION (3),CHLORIDE ION (2)	
3HPR_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE	
1E4V_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE	
5EJE_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION	
1E4Y_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE	
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION				
6HAP_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE	
6HAM_A				BIS(ADENOSINE)-5'-PENTAPHOSPHATE	
4K46_A				ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION	
3GMT_A				SULFATE ION (2)	
4PZL_A				CALCIUM ION,FORMIC ACID,GLYCEROL	

	source
1AKE_A	Escherichia coli
6S36_A	Escherichia coli
6RZE_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
3GMT_A	Burkholderia pseudomallei 1710b
4PZL_A	Francisella tularensis subsp. tularensis SCHU S4

1AKE_A	STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB
6S36_A	
6RZE_A	
3HPR_A	
1E4V_A	
5EJE_A	
1E4Y_A	
3X2S_A	
6HAP_A	
6HAM_A	
4K46_A	
3GMT_A	
4PZL_A	

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al.	J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al.	Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al.	Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al.	Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al.	Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al.	Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al.	Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al.	Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al.	To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al.	Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al.	To be published	0.19360	0.23680

rWork spaceGroup

```

1AKE_A 0.19600 P 21 2 21
6S36_A 0.15940 C 1 2 1
6RZE_A 0.18190 C 1 2 1
3HPR_A 0.20620 P 21 21 2
1E4V_A 0.19600 P 21 2 21
5EJE_A 0.18630 P 21 2 21
1E4Y_A 0.17800 P 1 21 1
3X2S_A 0.20700 P 21 21 21
6HAP_A 0.22370 I 2 2 2
6HAM_A 0.20311 P 43
4K46_A 0.16730 P 21 21 21
3GMT_A 0.23500 P 1 21 1
4PZL_A 0.19130 P 32

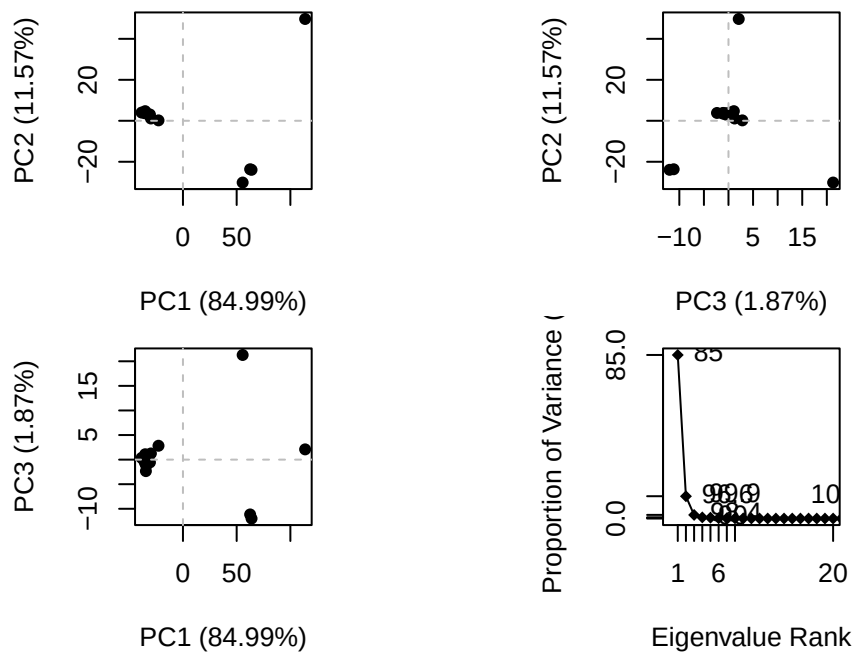
```

Principal component analysis

```

# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)

```



```

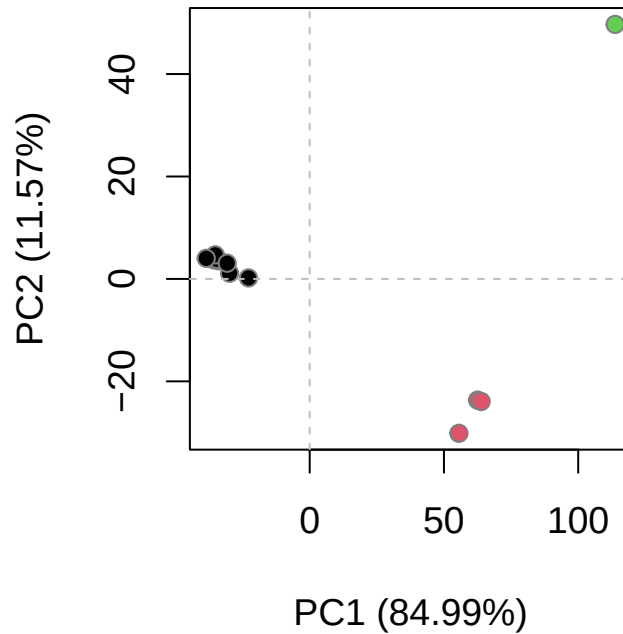
# Calculate RMSD
rd <- rmsd(pdbx)

```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



PCA Visualization

```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
```

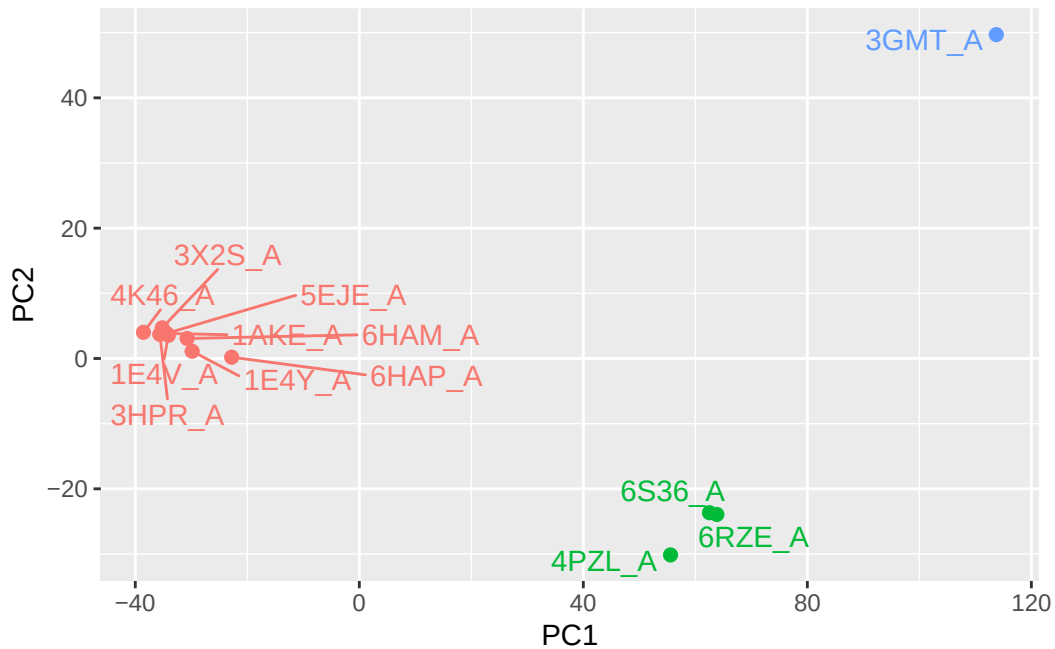


```

aes(PC1, PC2, col=col, label=ids) +
geom_point(size=2) +
geom_text_repel(max.overlaps = 20) +
theme(legend.position = "none")

```

p



Normal mode analysis

```

# NMA of all structures
modes <- nma(pdbbs)

```

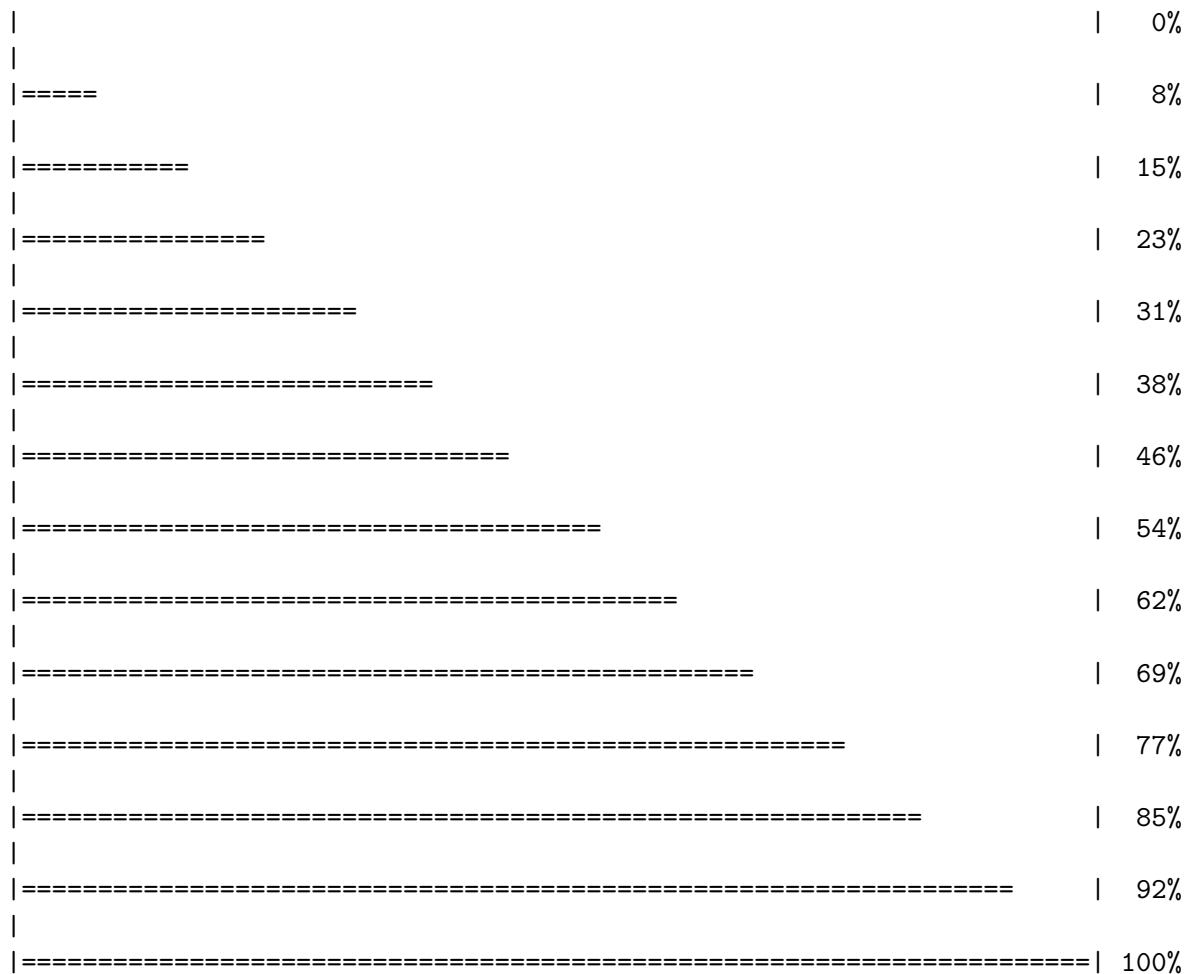
Details of Scheduled Calculation:

```

... 13 input structures
... storing 606 eigenvectors for each structure
... dimension of x$U.subspace: ( 612x606x13 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 36.9 Mb

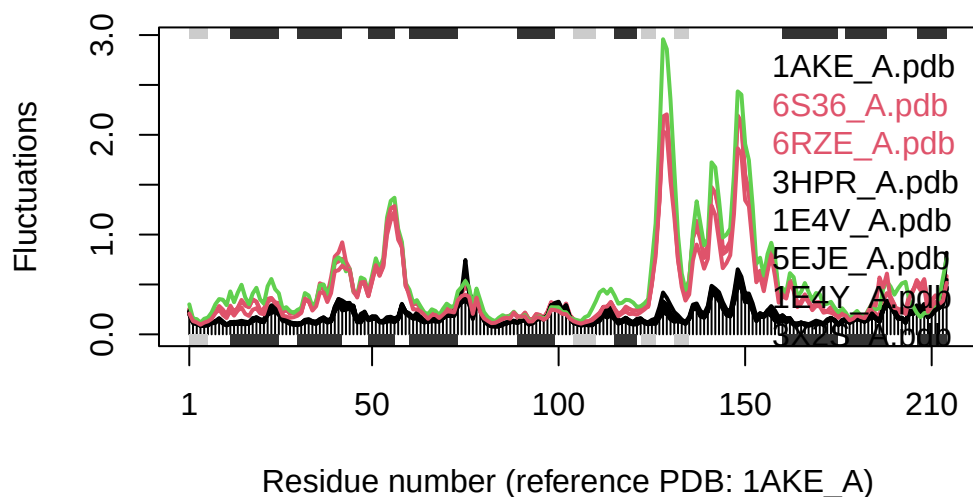
```

|



```
#Create plot based on the nma() function
plot(modes, pdbc, col=grps.rd)
```

Extracting SSE from pdbc\$sse attribute



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

Answer Q14: In this bio3d NMA plot, the colored lines is the flexibility profiles of groups and the black line is the flexibility profile from the reference model. The plot indicates the existence of two major distinct conformational states for Adk. The black and colored lines are relatively similar, as the peaks and valleys tend to occur at the same position along the sequence just at different amplitudes. The black and colored lines differ by a collective low frequency displacement of two nucleotide-binding site regions.